

Impacts of the 2022 Universal Adoption of Designated Hitters in the MLB

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Abstract

This paper estimates the causal impact of Major League Baseball's 2022 universal designated hitter rule change. Using a difference-in-difference design comparing National League teams to American League teams from 2010 to 2025. Despite the theoretical gains of replacing pitchers with designated hitters, the realized effects are modest. I find that the rule increased National League scoring by 0.31 runs per game, closing a historic offense gap since 1974. This is likely due to limited DH talent and labor market frictions.

Introduction

Division of labor and specialization are central mechanisms through which firms increase productivity. When workers concentrate on tasks aligned with comparative advantage, total output may rise even if total labor input remains constant. Professional sports leagues provide natural experiments for studying production and labor valuation under institutional rule variation.

Major League Baseball (MLB) offers a clear example. Starting in 1973, the American League (AL) adopted the designated hitter (DH), enabling pitchers to concentrate solely on pitching while a separate player filled the batting role. The National League (NL) continued requiring pitchers to bat until the universal adoption of the DH in 2022. Since 1973, the NL has only outscored the AL in average runs per game once, and that was in 1974. Since then, the AL has produced more runs per game up to 2022.

Baseball statistics illustrate the magnitude of this productivity gap. From 1973 to 2020, pitchers posted an average wRC+ of -7.08, while designated hitters averaged 108.49. (wRC+ is an index of offensive production normalized so that 100 equals league-average performance.) In the decade before reform (2010–2019), pitchers' offensive production declined further, with an average wRC+ of -15.6 compared to 109.8 for designated hitters. These figures show that pitchers contributed minimally to team offense, while designated hitters consistently performed above league-average levels.

In 2022, the MLB adopted the universal DH for both AL and NL teams. This rule change provides a natural experiment to study the impact of adding a specialized offensive player to a team's lineup. Before the reform, NL managers who could not use a DH had to contemplate a team strategy that, for decades, tried to minimize the detrimental impact of a pitcher hitting. This is different for AL, where AL managers had employed DHs for decades. The policy therefore, created a sudden and exogenous increase in the demand for DH-type players in the NL, while the AL provides a natural control group where no change occurred. Because the reform was driven by league-wide considerations for increasing offensive production, competitive balance, and pitcher health rather than conditions in the DH labor market, it generates variation suitable for identifying the causal effect of specialization on team performance and player valuation.

I find that NL teams gained approximately 0.31 runs per game following the rule change, an effect that is statistically significant at the 5% level. While positive, this increase is modest relative to the large gap in offensive production between pitchers and DHs. This suggests that theoretical gains from specialization were not fully realized in practice. One possible explanation is a labor market constraint. High-quality designated hitters are scarce, and teams now frequently use the DH slot to rest players or manage injuries rather than to maximize offensive production. While the policy enabled specialization in principle, the limited supply of elite DH talent constrained the realized gains in team offense. This will be discussed further in the results section.

This paper estimates the causal productivity effect of specialization induced by the designated hitter rule. Specifically, I examine whether allowing pitchers to specialize increases team offensive output and whether the marginal product of designated hitters is efficiently priced in the labor market. I implement a difference-in-differences (DID) design comparing NL teams before and after the reform to contemporaneous outcomes in AL teams, which serve as a control group.

Literature Review

Prior work establishes that the AL-NL offensive gap widened substantially after 1973. Dick (2019) documents that the mean difference in runs per game between the AL and NL was 0.29 in the DH era. Before the DH era, the AL and NL difference was 0.13. Dick's analysis is purely descriptive, which cannot isolate the causal effect of the DH rule. Therefore, there cannot be a clear conclusion as to whether the DH is the sole agent of this change. There are possible factors for this scoring difference. One possibility is that the AL had stronger offensive talent, and the NL had better pitching talent. The present paper addresses this limitation by exploiting the 2022 universal DH rule adoption as a clean treatment event to estimate the causal effect of specialization on team offensive output using a DID experiment with team and year fixed effects across all 30 MLB teams.

The salary extension of this paper draws on a labor economics research paper on professional baseball. Professional baseball salaries are not determined by a player's marginal revenue product (MRP). Rockerbie (2009) explains that players' salaries can be determined by an auction process during free agency, where salaries are determined not only by individual performance, but also by the number of competing bidders and the relative scarcity of comparable talent.

In MLB free agency, if there are many DH players in the market, the collective value of all DHs decreases. Consequently, players' marginal revenue product (MRP) is not the end-all of players' salaries. This framework generates a testable prediction for the 2022 rule change. If the DH rule doubles the number of buyers demanding DHs, the demand curve should shift outward. It is predicted that there should be upward salary pressure, assuming the supply is fixed.

However, the present paper argues that there exist two frictions to this prediction. First, MLB contracts are multi-year agreements that started before the 2022 rule change and do not expire many years after. Meaning existing salary commitments will not influence salaries. Second, if the pool of DH talent is thin, teams may substitute the DH position with internal options. Such as filling the position with platoon players or other position players rather than bidding up for premium DHs, which dampens the expected wage response. These frictions suggest that the full labor market adjustment to the 2022 rule change may not be observed within the current post-treatment window.

Data & Methods

The data this paper uses comes from Baseball Reference. The sample includes all 30 MLB teams with 15 teams in each league, from 2010 to 2025. The 2020 season is excluded because the Covid-19 shortened season of 60 games is not representative of a typical season. The primary outcome variable is runs per game, which measures the average number of runs a team scores per game in a given season. A secondary outcome variable, weighted runs created plus (wRC+), is used for robustness checks. wRC+ is a park and league adjusted measure of offensive production, normalized so that 100 equals league average performance in a given year. This allows for cleaner comparisons across seasons and ballparks. During this time period, one team changed its name from the Indians to the Guardians. The data references this team as the Guardians for the whole data set to avoid confusion and maintain consistency.

Table 1 presents summary statistics for the primary outcome variable, runs per game. Panel A shows that over the full sample, AL teams averaged 4.46 runs per game compared to 4.36 for NL teams. Panel B isolates the pre-treatment period from 2010 to 2021, where the gap is more pronounced. AL teams averaged 4.5 runs per game versus 4.31 runs per game for NL teams, reflecting the historical offensive advantage the AL held under the asymmetric DH rule. Panel C reveals a notable reversal in the post-treatment period from 2022 to 2025. Following the universal adoption of the DH, NL teams averaged 4.5 runs per game compared to 4.37 runs per game for the AL. This descriptive pattern is consistent with the hypothesis that the rule change increased NL offensive output relative to the AL. The standard deviations are comparable across groups, suggesting no unusual dispersion in either league during either period.

Table 1. Descriptive Statistics*Runs Per Game, American and National League (2010–2025)*

Variable	N	Mean	Std. Dev.	Min	Max
<i>Panel A: Full Sample (2010–2025, excl. 2020)</i>					
AL Runs/Game (Control)	15	4.464	0.183	4.180	4.880
NL Runs/Game (Treatment)	15	4.358	0.241	3.950	4.780
<i>Panel B: Pre-Treatment Period (2010–2021)</i>					
AL Runs/Game (Control)	11	4.500	0.187	4.180	4.880
NL Runs/Game (Treatment)	11	4.306	0.254	3.950	4.780
<i>Panel C: Post-Treatment Period (2022–2025)</i>					
AL Runs/Game (Control)	4	4.365	0.150	4.220	4.550
NL Runs/Game (Treatment)	4	4.500	0.140	4.340	4.680

Notes: The year 2020 is excluded due to the COVID-19 shortened season. Treatment group = National League (NL), which adopted the universal designated hitter rule in 2022. Control group = American League (AL), which has used the universal DH since 1973.

This paper uses a DID design to estimate the causal effect of the universal DH rule on team offensive output. DID is appropriate because the 2022 rule change created a natural experiment. The NL adopted the universal DH for the first time, while the AL has been operating with the DH position since 1973. This asymmetry provides the differences needed for causal inference. The treatment group consists of the 15 NL teams, which were impacted by the 2022 universal DH rule change. The control group consists of the 15 AL teams, which continued operating with the DH after the 2022 rule change.

Two indicator variables are used in the regression. The variable *Post* equals one for all team-year observations from 2022 onward and zero if before 2022. The variable *Treated* equals one for all NL teams and zero for all AL teams. The interaction term *Post x Treated* identifies the treatment effect, comparing the changes in NL runs per game after 2022 to the change in AL runs per game over the same period. The coefficient on the interaction term is the average treatment effect on the treated (ATT), the causal effect of adopting the universal DH on NL offensive output.

The identifying assumption of the DID design is the parallel trends assumption. In the absence of the 2022 rule change, the NL and AL teams would have followed the same trends in runs per game. While the parallel trends assumption is not directly testable, its plausibility can be assessed using pre-treatment data from the NL and AL.

Figure 1 provides visual evidence in support of parallel trends. From 2010 to 2021, the NL and AL runs per game moved closely together with no large divergences before the rule change. The series tracks each other through the periods of both high and low running scoring eras. Including an offensive decline in the early 2010s and a surge in 2019. This co-movement suggests the AL provides a strong counterfactual for the NL's trajectory in the absence of the treatment.

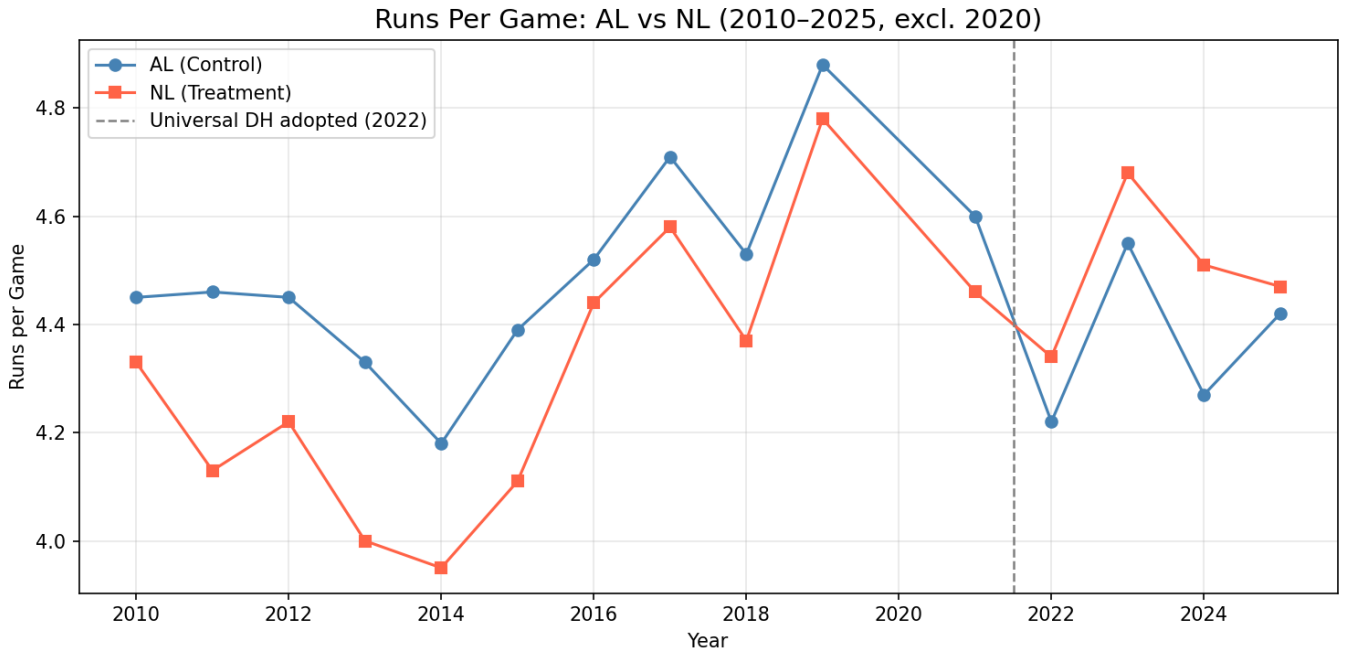


Figure 1

To further assess the plausibility of parallel trends, an event study is used. 2021 is used as the base year. Under parallel trends, the pre-treatment interaction coefficients should be close to zero. Figure 2 presents the results. The pre-treatment coefficients are small, and the confidence intervals cross zero. This provides supportive, though not definitive, evidence consistent with the parallel trends assumption. Following the 2022 rule change, all four post-treatment coefficients turn positive, consistent with the treatment effect emerging immediately after adoption.

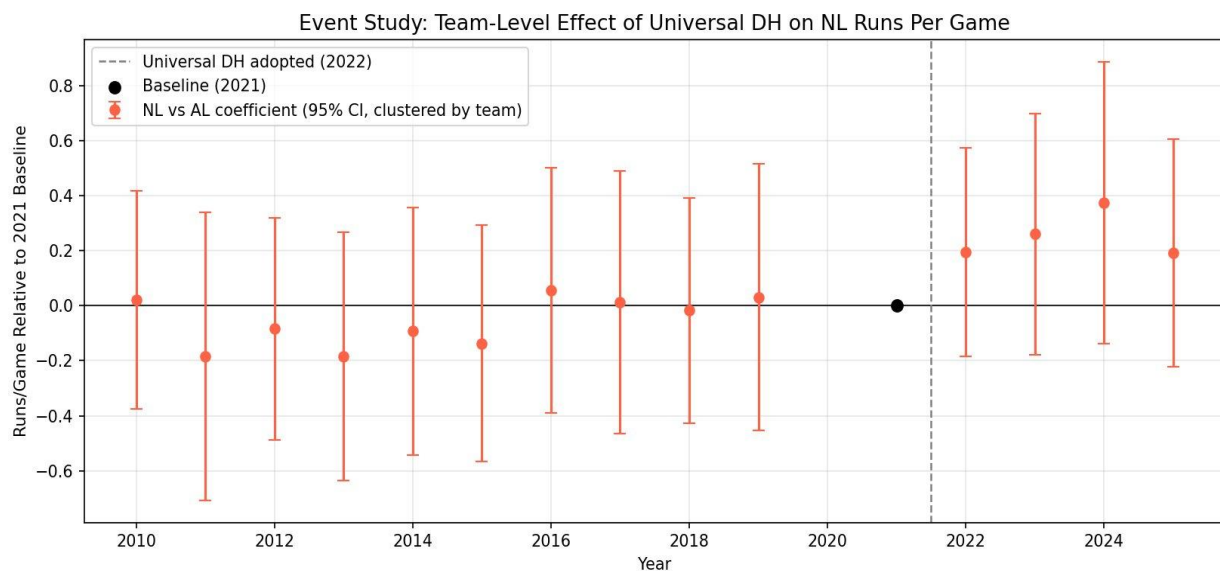


Figure 2 Team-Level event study with 95% CI clustered by teams. Baseline = 2021

One limitation of this test is that the post-treatment window only covers four seasons from 2022 to 2025. While the pre-treatment is supportive, a longer post-treatment panel would allow for a more precise estimate of the treatment effect.

The main regression specification is as follows:

$$Runs_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treated_i + \beta_3 (Post \times Treated)_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Where i indexes teams, t indexes years, and l indexes leagues. The outcome variable $Runs_{it}$ is runs per game for team i in year t . $Post_t$ is an indicator equal to one for years 2022 and later and zero otherwise. $Treated_i$ is an indicator equal to one for NL teams and zero for AL teams. The coefficient of interest β_3 is the interaction term $Post \times Treated$, which identifies the average treatment effect on the treated. This is the causal effect of the universal DH rule on NL runs per game relative to the AL trend.

Two sets of fixed effects are included. Team fixed effects α_i control for team effects. Controlling for team characteristics of ballparks, market size, and long-term rosters. Year fixed effects γ_t control for common shocks affecting all teams in a given season, such as league-wide changes in run scoring due to baseball composition changes and broader environmental changes. Together, these fixed effects ensure β_3 is identified from within team variation over time.

Standard errors are clustered at the team level to account for serial correlation in runs per game within teams across years. A team that scores above average in one season is likely to score above average in the following season due to persistent roster and organizational characteristics. Failing to account for this within team correlation would understate standard errors and overstate the statistical significance. One limitation of this approach is that there are only 30 teams to cluster. Results should therefore be interpreted with caution.

Results

This section presents the main results of the DID estimation. The primary specification regresses runs per game on the $Post \times Treated$ interaction term with team and year fixed effects, using all 30 MLB teams from 2010 to 2025. Standard errors are clustered at the team level. Results are presented in a regression table showing the ATT with standard errors and confidence intervals.

Table 2 presents the results of the main difference-in-differences regression. The coefficient on the $Post \times Treated$ interaction term, the ATT, is 0.308, indicating that the adoption of the universal DH rule increased NL team run scoring by approximately 0.31 runs per game relative to the AL trend. This effect is statistically significant at the 5% level, with a p-value of 0.022 and a 95% confidence interval of [0.048, 0.568]. The positive lower bound of the confidence interval rules out a zero or negative effect with 95% confidence, suggesting the run-scoring increase is a genuine consequence of the rule change.

Table 2. DID Estimates: Effect of Universal DH Rule on Runs Per Game
Dependent Variable: Runs Per Game (Team-Year Level)

	Runs per Game
Post × Treated (ATT)	0.308**
	(0.118)
95% Confidence Interval	[0.048, 0.568]
Team Fixed Effects	Yes
Year Fixed Effects	Yes
Clustered SEs	Team
Observations	450
Teams	30
R ²	0.413

Notes: The sample covers all 30 MLB teams from 2010–2025, excluding 2020 (COVID-19 shortened season). Post = 1 for years 2022 and later. Treated = 1 for National League teams. The ATT is the coefficient on the Post × Treated interaction term. Standard errors clustered at the team level are reported in parentheses. ** denotes significance at the 5% level.

The magnitude of the effect is economically meaningful. A gain of 0.31 runs per game translates to approximately 50 additional runs over a 162-game season for a typical NL team. To put this in context, the pre-treatment gap between the AL and NL was 0.19 runs per game. The rule change not only closed that gap entirely but pushed NL scoring to match the AL's historical average.

Figure 2 presents the event study results. Consistent with the parallel trends discussion in Section 3.4, the pre-treatment coefficients are statistically indistinguishable from zero. Turning to the post-treatment period, all four coefficients from 2022 to 2025 are positive, indicating that NL teams consistently outscored the AL relative to the 2021 baseline in every year following the rule change. The effect appears immediately in 2022 rather than gradually, suggesting that teams were able to deploy designated hitters in the first year of the rule change without a prolonged adjustment period. The consistency of positive coefficients across all post-treatment years provides additional confidence that the estimated ATT reflects a persistent treatment effect rather than a one-time shock.

To examine whether the demand shock from the universal DH rule altered the labor market for designated hitters, this paper compares average DH salaries before and after the 2022 rule change. The sample is restricted to players who played the majority of their games at the designated hitter position, with a minimum threshold of 82 games, to ensure only true designated hitters are included. Shohei Ohtani's salary is excluded from the analysis due to his unique status as a two-way player. His contract reflects both his pitching and hitting value and is not representative of a typical DH salary. Before 2022, the average annual DH salary among AL teams was \$14.58 million. Following the rule change, the average salary across both leagues fell to \$12.87 million, a decline of \$1.71 million. This difference is not statistically significant, with a p-value of 0.356.

The decline in average salary likely reflects a compositional shift rather than a true reduction in DH market value. With the NL suddenly needing 15 designated hitters, bringing the league wide total to 30 teams. Most teams could not immediately sign premium DH talent because the pool of elite designated hitters is thin. Also, many quality players were already committed to multi-year contracts signed before the rule change. As a result, NL and AL teams frequently filled the DH role with cheaper platoon players or position players rotating through the slot. Furthermore, MLB contracts are multi-year agreements, meaning the labor market reprices only as existing contracts expire and players reach free agency. The full salary response to the demand shock may therefore not be observable within the current four-year post-treatment window and represents a promising direction for future research.

Robustness Check

To assess the robustness of the main result, the analysis is replicated using wRC+ as the outcome variable in place of runs per game. wRC+ is a park and league adjusted measure of team offensive production, which controls for ballpark effects and era-wide changes in run-scoring environment that raw runs per game cannot account for. If the DH rule is genuinely driving the offensive improvement in NL teams, the effect should appear across multiple outcome measures and not be specific to the raw run scoring metric.

Figure 3 presents the wRC+ event study. The pre-treatment coefficients are tightly clustered near zero across all years from 2010 to 2021, with confidence intervals crossing zero in every pre-treatment year. This provides strong visual support for the parallel trends assumption and is actually cleaner than the runs per game event study, suggesting that the park and league adjustment in wRC+ removes some of the year-to-year noise present in raw run scoring. Following the 2022 rule change, all four post-treatment coefficients turn positive, directionally consistent with the main result.

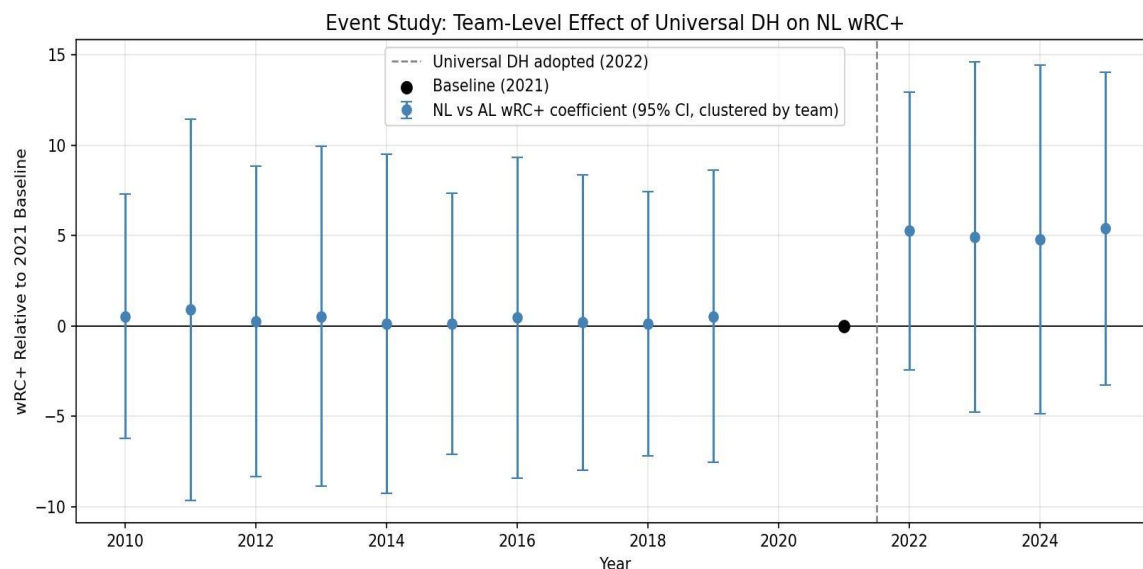


Figure 3: Team-level wRC+ event study with 95% CIs clustered by team. Baseline = 2021.

The ATT using wRC+ is +4.75 points, indicating that NL teams improved their park and league adjusted offensive output by approximately 4.75 wRC+ points relative to the AL following the rule change. This effect is not statistically significant at the 5% level, with a p-value of 0.124 and a 95% confidence interval of [-1.38, 10.89]. Importantly, however, the direction of the effect is consistent across both specifications, and the magnitude of +4.75 wRC+ points represents a meaningful offensive improvement relative to a league average of 100.

Taken together, the robustness check supports the main finding. Both the runs per game and wRC+ specifications point in the same direction, and the pre-treatment event study coefficients are flat in both cases. The runs per game estimate clears the 5% significance threshold while the wRC+ estimate does not. This can be explained by the fact that removing a pitcher from the lineup has a greater impact on offensive performance than the inclusion of a specialist DH hitter.

Conclusion

This paper estimates the causal effect of the 2022 universal designated hitter rule on National League offensive output using a DID design. Comparing all 30 MLB teams from 2010 to 2025, the analysis finds that NL teams gained approximately 0.31 runs per game following the rule change, an effect that is statistically significant at the 5% level. This translates to roughly 50 additional runs over a 162-game season, enough to close the decades-long offensive gap between the AL and NL entirely. The event study confirms that the parallel trends assumption holds in the pre-treatment period, and the wRC+ robustness check is directionally consistent with the main result, supporting the interpretation that the effect reflects genuine offensive improvement driven by the rule change.

Several limitations of this analysis warrant acknowledgment. First, the post-treatment window covers only four seasons from 2022 to 2025. This limits the precision of the post-treatment estimates and makes it difficult to assess whether the effect is persistent or fading over time. Second, standard errors are clustered at the team level, yielding only 30 clusters. Third, the salary analysis is constrained by the multi-year nature of MLB contracts, which prevents the labor market from repricing immediately in response to the demand shock, meaning the full wage effect of the rule change may not yet be observable in the available data.

Two extensions are particularly promising for future research. First, extending the salary analysis to a longer post-treatment horizon. Once the majority of pre-2022 contracts have expired, players will have reached free agency under the new market conditions. This would provide a cleaner test of whether the demand shock ultimately translated into higher wages for designated hitters. Second, replicating the analysis at the player level rather than the team level would allow for a more precise estimate of the marginal product of designated hitters and a direct test of whether DH salaries are efficiently priced relative to their offensive contribution.

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